

HUMAN ANATOMY AND PHYSIOLOGY(2) A+ MCQ'S TRAILS

1. Urinary System

1. What is the main functional unit of the kidney?

- A) Nephron
- B) Glomerulus
- C) Renal pelvis
- D) Ureter

Answer: A) Nephron

2. Which hormone is primarily responsible for regulating sodium and water balance in the body?

- A) Aldosterone
- B) Antidiuretic hormone (ADH)
- C) Cortisol
- D) Renin

Answer: A) Aldosterone

3. What is the primary role of the renal corpuscle?

- A) Filtration of blood
- B) Reabsorption of nutrients
- C) Secretion of hormones
- D) Regulation of blood pressure

Answer: A) Filtration of blood

4. Which structure of the nephron is responsible for reabsorbing water and solutes into the blood?

- A) Proximal convoluted tubule
- B) Loop of Henle
- C) Distal convoluted tubule
- D) Collecting duct

Answer: A) Proximal convoluted tubule

5. What is the primary function of the bladder?

- A) Filtration of waste
- B) Storage of urine
- C) Regulation of pH

- D) Production of urine

Answer: B) Storage of urine

6. The process of urine formation occurs through which three processes?

- A) Filtration, reabsorption, secretion
- B) Secretion, filtration, excretion
- C) Absorption, filtration, secretion
- D) Excretion, secretion, reabsorption

Answer: A) Filtration, reabsorption, secretion

7. Where are the kidneys located in the human body?

- A) Pelvic cavity
- B) Peritoneal cavity
- C) Retroperitoneal space
- D) Abdominal cavity

Answer: C) Retroperitoneal space

8. Which type of epithelium lines the renal tubules?

- A) Simple squamous epithelium
- B) Transitional epithelium
- C) Simple cuboidal epithelium
- D) Stratified squamous epithelium

Answer: C) Simple cuboidal epithelium

9. What structure carries urine from the kidney to the urinary bladder?

- A) Urethra
- B) Ureter
- C) Renal pelvis
- D) Nephron

Answer: B) Ureter

10. Which part of the nephron is primarily involved in the reabsorption of water?

- A) Distal convoluted tubule
- B) Loop of Henle
- C) Proximal convoluted tubule
- D) Collecting duct

Answer: D) Collecting duct

2. Reproductive System

11. What is the primary male reproductive hormone?

- A) Testosterone
- B) Estrogen
- C) Progesterone
- D) Luteinizing hormone

Answer: A) Testosterone

12. Which structure in the female reproductive system is responsible for the production of oocytes?

- A) Uterus
- B) Ovaries
- C) Fallopian tubes
- D) Vagina

Answer: B) Ovaries

13. At which stage of meiosis do homologous chromosomes separate?

- A) Prophase I
- B) Metaphase I
- C) Anaphase I
- D) Telophase II

Answer: C) Anaphase I

14. What is the function of the fallopian tubes?

- A) Nourish the embryo
- B) Transport sperm
- C) Site of fertilization
- D) Produce hormones

Answer: C) Site of fertilization

15. Which hormone is responsible for triggering ovulation?

- A) Follicle-stimulating hormone (FSH)
- B) Luteinizing hormone (LH)
- C) Estrogen
- D) Progesterone

Answer: B) Luteinizing hormone (LH)

16. What is the function of the scrotum?

- A) Protecting the testes from injury

- B) Regulating the temperature of the testes
- C) Transporting sperm
- D) Producing testosterone

Answer: B) Regulating the temperature of the testes

17. What is the primary function of the prostate gland?

- A) Produce sperm
- B) Produce seminal fluid that nourishes and transports sperm
- C) Store sperm
- D) Regulate hormone levels

Answer: B) Produce seminal fluid that nourishes and transports sperm

18. Which structure connects the ovaries to the uterus?

- A) Uterine wall
- B) Vagina
- C) Fallopian tubes
- D) Cervix

Answer: C) Fallopian tubes

19. What is the function of the ovary?

- A) To produce estrogen and progesterone
- B) To facilitate sperm transport
- C) To provide a site for fertilization
- D) To support fetal development

Answer: A) To produce estrogen and progesterone

20. Which of the following is the term for the monthly cycle of changes in the female reproductive system?

- A) Gestation
- B) Menstrual cycle
- C) Fertilization
- D) Ovulation

Answer: B) Menstrual cycle

21. Which cells in the testes are responsible for testosterone production?

- A) Sertoli cells
- B) Leydig cells
- C) Spermatogonia
- D) Germ cells

Answer: B) Leydig cells

22. What occurs during the luteal phase of the menstrual cycle?

- A) Follicle maturation
- B) Ovulation
- C) Corpus luteum formation
- D) Menstruation

Answer: C) Corpus luteum formation

23. What allows the sperm to penetrate the ovum?

- A) Acrosome reaction
- B) Fertilization membrane
- C) Zona pellucida
- D) Cortical granules

Answer: A) Acrosome reaction

24. During which phase does the endometrium thicken in preparation for potential implantation?

- A) Follicular phase
- B) Luteal phase
- C) Menstrual phase
- D) Ovulatory phase

Answer: B) Luteal phase

25. What is the primary function of the vagina?

- A) Site of sperm deposit
- B) Birth canal
- C) Passage for menstrual flow
- D) All of the above

Answer: D) All of the above

3. Endocrine System

26. What is the primary hormone produced by the pancreas that regulates blood glucose levels?

- A) Insulin
- B) Glucagon
- C) Somatostatin
- D) Cortisol

Answer: A) Insulin

27. Which gland is often referred to as the "master gland" of the endocrine system?

- A) Thyroid gland
- B) Adrenal gland
- C) Pituitary gland
- D) Pineal gland

Answer: C) Pituitary gland

28. What is the function of thyroid hormones?

- A) Regulation of sleep cycles
- B) Regulation of metabolic rate
- C) Regulation of blood sugar levels
- D) Regulation of blood pressure

Answer: B) Regulation of metabolic rate

29. Which hormone is produced by the adrenal cortex?

- A) Epinephrine
- B) Norepinephrine
- C) Cortisol
- D) Insulin

Answer: C) Cortisol

30. What regulates the secretion of hormones from the anterior pituitary gland?

- A) Direct neural connection
- B) Blood-borne

30. What regulates the secretion of hormones from the anterior pituitary gland?

- A) Direct neural connection
- B) Blood-borne hormones from the hypothalamus
- C) Feedback inhibition from peripheral hormones
- D) Nerve impulses from the spinal cord

Answer: B) Blood-borne hormones from the hypothalamus

31. Which hormone is secreted by the adrenal medulla?

- A) Cortisol
- B) Aldosterone
- C) Epinephrine
- D) Insulin

Answer: C) Epinephrine

32. What is the role of parathyroid hormone?

- A) Decrease calcium levels in the blood
- B) Increase calcium levels in the blood
- C) Control blood glucose levels
- D) Regulate the metabolism of fats

Answer: B) Increase calcium levels in the blood

33. Which gland is responsible for producing oxytocin and vasopressin (ADH)?

- A) Thyroid gland
- B) Pituitary gland
- C) Hypothalamus
- D) Pancreas

Answer: C) Hypothalamus

34. What is the primary effect of growth hormone (GH)?

- A) Increase blood glucose levels
- B) Promote tissue growth and repair
- C) Regulate water balance
- D) Control metabolism

Answer: B) Promote tissue growth and repair

35. Which condition is caused by an overproduction of growth hormone in adults?

- A) Dwarfism
- B) Acromegaly
- C) Gigantism
- D) Addison's disease

Answer: B) Acromegaly

36. What effect does insulin have on blood glucose levels?

- A) Increases blood glucose levels
- B) Decreases blood glucose levels
- C) Stabilizes blood glucose levels
- D) Causes glucose to be stored as fat

Answer: B) Decreases blood glucose levels

37. Which hormone primarily stimulates the release of glucose from the liver?

- A) Insulin
- B) Glucagon
- C) Somatostatin

- D) Cortisol

Answer: B) Glucagon

38. Which gland produces the hormone melatonin?

- A) Thyroid gland
- B) Pituitary gland
- C) Pineal gland
- D) Adrenal gland

Answer: C) Pineal gland

39. What is the psychological effect of cortisol during stress?

- A) Decrease in energy levels
- B) Increase in immune function
- C) Increase in blood glucose levels
- D) Stabilizes mood

Answer: C) Increase in blood glucose levels

40. Which hormone promotes water reabsorption by the kidneys?

- A) Aldosterone
- B) Antidiuretic hormone (ADH)
- C) Renin
- D) Insulin

Answer: B) Antidiuretic hormone (ADH)

4. Nervous System

41. What is the primary function of neurons?

- A) Provide support to the nervous tissue
- B) Carry electrical impulses
- C) Produce neurotransmitters
- D) Maintain homeostasis

Answer: B) Carry electrical impulses

42. Which part of the neuron receives signals from other neurons?

- A) Axon
- B) Soma
- C) Dendrites
- D) Myelin sheath

Answer: C) Dendrites

43. What is the role of the myelin sheath?

- A) Protects the neuron from damage
- B) Insulates the axon to speed up transmission
- C) Produces neurotransmitters
- D) Receives signals from other neurons

Answer: B) Insulates the axon to speed up transmission

44. Which neurotransmitter is primarily involved in mood regulation?

- A) Acetylcholine
- B) Dopamine
- C) Norepinephrine
- D) Serotonin

Answer: D) Serotonin

45. What is the primary function of the spinal cord?

- A) Information storage
- B) Relay signals between the brain and the rest of the body
- C) Produce neurotransmitters
- D) Process sensory input

Answer: B) Relay signals between the brain and the rest of the body

46. Which part of the brain is responsible for coordination and balance?

- A) Cerebrum
- B) Cerebellum
- C) Brainstem
- D) Thalamus

Answer: B) Cerebellum

47. What is the main function of the autonomic nervous system?

- A) Controls voluntary muscle movements
- B) Regulates involuntary bodily functions
- C) Processes sensory information
- D) Coordinates reflex actions

Answer: B) Regulates involuntary bodily functions

48. Which structure is responsible for the blood-brain barrier?

- A) Astrocytes
- B) Oligodendrocytes
- C) Schwann cells

- D) Microglia
- Answer: A) Astrocytes**

49. What type of glial cells are responsible for the production of myelin in the central nervous system?

- A) Schwann cells
 - B) Oligodendrocytes
 - C) Astrocytes
 - D) Ependymal cells
- Answer: B) Oligodendrocytes**

50. Which cranial nerve is responsible for vision?

- A) Olfactory nerve
 - B) Optic nerve
 - C) Oculomotor nerve
 - D) Trochlear nerve
- Answer: B) Optic nerve**

51. What is the function of the hypothalamus?

- A) Regulates body temperature and thirst
 - B) Controls heart rate
 - C) Processes sensory information
 - D) Coordinates voluntary movement
- Answer: A) Regulates body temperature and thirst**

52. What does the term "reflex arc" refer to?

- A) A pathway for voluntary movements
 - B) A neural pathway that controls reflex actions
 - C) A mechanism for sensory processing
 - D) A part of the cerebral cortex
- Answer: B) A neural pathway that controls reflex actions**

53. Which type of neurotransmitter is associated with muscle contraction?

- A) Acetylcholine
 - B) Serotonin
 - C) Dopamine
 - D) Norepinephrine
- Answer: A) Acetylcholine**

54. Which part of the brain regulates basic life functions such as heart rate and breathing?

- A) Cerebellum
- B) Medulla oblongata
- C) Cerebrum
- D) Thalamus

Answer: B) Medulla oblongata

55. What is the primary role of the limbic system?

- A) Coordination and balance
- B) Emotions and memory
- C) Sensory processing
- D) Motor control

Answer: B) Emotions and memory

56. What kind of information is transmitted by afferent neurons?

- A) Motor signals from the brain to muscles
- B) Sensory information from the body to the brain
- C) Reflex signals
- D) None of the above

Answer: B) Sensory information from the body to the brain

57. Which of the following structures is part of the peripheral nervous system?

- A) Spinal cord
- B) Brain
- C) Cranial nerves
- D) Cerebellum

Answer: C) Cranial nerves

58. What is the function of the sensory (afferent) division of the peripheral nervous system?

- A) Controls voluntary muscles
- B) Transmits sensory information to the brain
- C) Regulates heart rate
- D) Coordinates reflexes

Answer: B) Transmits sensory information to the brain

59. What is the primary effect of the sympathetic nervous system?

- A) Decrease heart rate
- B) Constrict pupils
- C) Dilate bronchi

- D) Increase digestion

Answer: C) Dilate bronchi

60. What is neuroplasticity?

- A) The brain's ability to grow new neurons
- B) The ability of the nervous system to adapt and change
- C) The repair of damaged nerves
- D) The process of neurotransmitter release

Answer: B) The ability of the nervous system to adapt and change

61. Which neurotransmitter is primarily involved in the reward system of the brain?

- A) Serotonin
- B) Norepinephrine
- C) Dopamine
- D) Acetylcholine

Answer: C) Dopamine

62. What is the primary function of the frontal lobe?

- A) Processing sensory information
- B) Planning and executing movements
- C) Memory storage
- D) Regulation of emotions

Answer: B) Planning and executing movements

63. What type of sensory receptors detect light?

- A) Chemoreceptors
- B) Photoreceptors
- C) Mechanoreceptors
- D) Thermoreceptors

Answer: B) Photoreceptors

64. Which part of the neuron is primarily responsible for transmitting impulses away from the cell body?

- A) Dendrite
- B) Synapse
- C) Axon
- D) Soma

Answer: C) Axon

65. What is the function of the somatic nervous system?

- A) Controls involuntary actions
- B) Regulates smooth and cardiac muscle
- C) Controls voluntary movement of skeletal muscles
- D) Processes sensory inputs

Answer: C) Controls voluntary movement of skeletal muscles

66. Which structure in the brain is responsible for regulating sleep-wake cycles?

- A) Thalamus
- B) Hypothalamus
- C) Cerebellum
- D) Medulla oblongata

Answer: B) Hypothalamus

67. What is a potential effect of a damaged spinal cord?

- A) Enhanced reflex responses
- B) Loss of sensation and motor control below the injury
- C) Increased sensitivity to pain
- D) Improved coordination

Answer: B) Loss of sensation and motor control below the injury

68. Which cranial nerve is primarily responsible for facial movement and expression?

- A) Trigeminal nerve
- B) Facial nerve
- C) Glossopharyngeal nerve
- D) Vagus nerve

Answer: B) Facial nerve

69. What is the role of the thalamus?

- A) Regulation of body temperature
- B) Relay station for sensory information
- C) Coordination of movement
- D) Control of autonomic functions

Answer: B) Relay station for sensory information

70. Which brain structure is involved in forming new memories?

- A) Amygdala
- B) Hippocampus
- C) Cerebellum
- D) Brainstem

Answer: B) Hippocampus

5. Lymphatic System

71. What is the primary function of the lymphatic system?

- A) Circulation of blood
- B) Defense against infection and disease
- C) Transportation of oxygen
- D) Regulation of body temperature

Answer: B) Defense against infection and disease

72. Which of the following is a primary lymphoid organ?

- A) Lymph nodes
- B) Spleen
- C) Thymus
- D) Appendix

Answer: C) Thymus

73. What is the role of lymph nodes?

- A) Production of red blood cells
- B) Filtering lymph and housing immune cells
- C) Storage of platelets
- D) Absorption of nutrients

Answer: B) Filtering lymph and housing immune cells

74. Which cells are primarily responsible for the immune response?

- A) Erythrocytes
- B) Thrombocytes
- C) Lymphocytes
- D) Neutrophils

Answer: C) Lymphocytes

75. What type of immunity involves the production of antibodies?

- A) Innate immunity
- B) Passive immunity
- C) Adaptive (acquired) immunity
- D) Natural immunity

Answer: C) Adaptive (acquired) immunity

76. Which type of lymphocyte is responsible for cell-mediated immunity?

- A) B cells
- B) T cells

- C) Natural killer cells
- D) Macrophages

Answer: B) T cells

77. Which lymphatic organ is involved in the recycling of red blood cells?

- A) Lymph nodes
- B) Thymus
- C) Spleen
- D) Bone marrow

Answer: C) Spleen

78. What is the major function of the spleen?

- A) Absorb fat
- B) Filter blood and produce lymphocytes
- C) Produce hormones
- D) Store bile

Answer: B) Filter blood and produce lymphocytes

79. What is edema in the context of the lymphatic system?

- A) Infection of lymph nodes
- B) The accumulation of lymph fluid in tissues
- C) Decreased blood flow to tissues
- D) An increase in blood protein levels

Answer: B) The accumulation of lymph fluid in tissues

80. Which structure collects lymph from the lower half of the body?

- A) Right lymphatic duct
- B) Thoracic duct
- C) Subclavian vein
- D) Jugular vein

Answer: B) Thoracic duct

81. What is the primary role of macrophages in the lymphatic system?

- A) Produce antibodies
- B) Phagocytize pathogens and debris
- C) Initiate the clotting process
- D) Transport lymph

Answer: B) Phagocytize pathogens and debris

82. Which of the following is NOT a function of the lymphatic system?

- A) Removal of interstitial fluid
- B) Absorption of dietary fats
- C) Regulation of blood pH
- D) Immune response

Answer: C) Regulation of blood pH

83. What is the term for lymphatic tissue located in the throat?

- A) Thymus
- B) Palatine tonsils
- C) Cervical lymph nodes
- D) Spleen

Answer: B) Palatine tonsils

84. Which of the following occurs during lymphatic drainage?

- A) Lymph is produced in the blood vessels.
- B) Lymph flows only in one direction toward the heart.
- C) Lymph can re-enter the bloodstream through arteries.
- D) Lymphatic vessels are under pressure similar to arteries.

Answer: B) Lymph flows only in one direction toward the heart.

85. What is the role of dendritic cells in the lymphatic system?

- A) Destroy infected cells
- B) Present antigens to T cells
- C) Produce antibodies
- D) Secrete lymph fluid

Answer: B) Present antigens to T cells

6. Muscles

86. What type of muscle is responsible for voluntary movements?

- A) Cardiac muscle
- B) Smooth muscle
- C) Skeletal muscle
- D) Involuntary muscle

Answer: C) Skeletal muscle

87. Which type of muscle tissue is striated and under voluntary control?

- A) Cardiac muscle
- B) Smooth muscle
- C) Skeletal muscle

- D) All muscle types
- Answer: C) Skeletal muscle**

88. Where is cardiac muscle found?

- A) In the walls of the heart
 - B) In the digestive tract
 - C) In the brain
 - D) In voluntary muscles
- Answer: A) In the walls of the heart**

89. What is the contractile unit of a muscle fiber called?

- A) Myofilament
 - B) Sarcomere
 - C) Myofibril
 - D) Muscle fiber
- Answer: B) Sarcomere**

90. Which protein is primarily responsible for muscle contraction?

- A) Collagen
 - B) Actin
 - C) Elastin
 - D) Keratin
- Answer: B) Actin**

91. What is the function of myosin in muscle cells?

- A) Store calcium ions
 - B) Provide structural support
 - C) Facilitate muscle contraction through cross-bridge cycling
 - D) Act as an energy source
- Answer: C) Facilitate muscle contraction through cross-bridge cycling**

92. What is the role of ATP in muscle contraction?

- A) It initiates muscle fiber repair.
 - B) It provides energy for muscle contraction and relaxation.
 - C) It transports oxygen to the muscles.
 - D) It inhibits muscle fatigue.
- Answer: B) It provides energy for muscle contraction and relaxation.**

93. Which type of muscle fibers are more resistant to fatigue?

- A) Fast-twitch fibers
 - B) Slow-twitch fibers
 - C) Intermediate fibers
 - D) None of the above
- Answer: B) Slow-twitch fibers**

94. Which type of muscle contraction occurs without a change in muscle length?

- A) Isotonic contraction
 - B) Isometric contraction
 - C) Concentric contraction
 - D) Eccentric contraction
- Answer: B) Isometric contraction**

95. What occurs during an eccentric contraction?

- A) The muscle shortens while generating force.
 - B) The muscle lengthens while generating force.
 - C) The muscle remains the same length while generating force.
 - D) The muscle relaxes completely.
- Answer: B) The muscle lengthens while generating force.**

96. What is the name of the connective tissue that surrounds an entire muscle?

- A) Endomysium
 - B) Perimysium
 - C) Epimysium
 - D) Sarcoplasm
- Answer: C) Epimysium**

97. What neurotransmitter is released at the neuromuscular junction to initiate muscle contraction?

- A) Norepinephrine
 - B) Acetylcholine
 - C) Serotonin
 - D) Dopamine
- Answer: B) Acetylcholine**

98. Which structure in muscle cells stores calcium ions necessary for contraction?

- A) Mitochondria
- B) Sarcoplasmic reticulum
- C) Golgi apparatus

- D) Lysosomes
- Answer: B) Sarcoplasmic reticulum**

99. What is the primary source of energy for muscle contraction during prolonged exercise?

- A) Creatine phosphate
- B) Glycogen
- C) Fatty acids
- D) Glucose

Answer: C) Fatty acids

100. Which of the following factors can influence muscle contraction strength?

- A) The number of muscle fibers recruited
- B) Frequency of stimulation
- C) Muscle length at rest
- D) All of the above

Answer: D) All of the above

101. What is hypertrophy in the context of muscle physiology?

- A) A decrease in muscle size
- B) An increase in muscle fiber number
- C) An increase in muscle fiber size
- D) A condition of muscle weakness

Answer: C) An increase in muscle fiber size

102. How do tendons function in the muscular system?

- A) Store energy for muscle contractions
- B) Connect muscles to bones
- C) Provide structural support
- D) Facilitate nerve signal transmission

Answer: B) Connect muscles to bones

103. What is the primary role of smooth muscle tissue?

- A) To facilitate voluntary movement
- B) To contract and propel substances through hollow organs
- C) To maintain posture
- D) To produce body heat

Answer: B) To contract and propel substances through hollow organs

104. Which part of the brain is primarily responsible for coordinating muscle movements?

- A) Cerebrum
- B) Cerebellum
- C) Brainstem
- D) Thalamus

Answer: B) Cerebellum

6. Muscles (Continued)

105. Which type of muscle is found in the walls of blood vessels and the digestive tract?

- A) Skeletal muscle
- B) Cardiac muscle
- C) Smooth muscle
- D) None of the above

Answer: C) Smooth muscle

106. What type of contraction occurs when a muscle shortens while producing tension?

- A) Isometric contraction
- B) Eccentric contraction
- C) Concentric contraction
- D) Isotonic contraction

Answer: C) Concentric contraction

107. Which muscle is responsible for raising the eyebrows?

- A) Orbicularis oris
- B) Temporalis
- C) Frontalis
- D) Zygomaticus

Answer: C) Frontalis

108. What is the term for the repetitive cycles of muscle contraction and relaxation?

- A) Tetanus
- B) Twitch
- C) Muscle tone
- D) Motor unit

Answer: B) Twitch

109. Which muscle type exhibits intercalated discs?

- A) Skeletal muscle
- B) Smooth muscle
- C) Cardiac muscle
- D) All muscle types

Answer: C) Cardiac muscle

110. Which process provides the initial energy source for muscle contraction?

- A) Aerobic respiration
- B) Anaerobic glycolysis
- C) ATP-Creatine phosphate system
- D) Beta-oxidation

Answer: C) ATP-Creatine phosphate system

111. **What is the function of the muscle spindle?**

- A) It stimulates muscle contraction.
- B) It senses muscle stretch and relays information to the nervous system.
- C) It connects muscles to bones.
- D) It stores energy for muscle contraction.

Answer: B) It senses muscle stretch and relays information to the nervous system.

112. **During which stage of muscle contraction does the sarcomere shorten?**

- A) Relaxation
- B) Contraction
- C) Recruitment
- D) Isotonic

Answer: B) Contraction

113. **What is the role of lactic acid in muscle physiology?**

- A) It increases muscle strength.
- B) It is a byproduct of anaerobic metabolism that can lead to muscle fatigue.
- C) It provides immediate energy for muscle contraction.
- D) It helps in muscle recovery post-exercise.

Answer: B) It is a byproduct of anaerobic metabolism that can lead to muscle fatigue.

114. **Which of the following best describes muscle tone?**

- A) A state of complete muscle relaxation
- B) The constant, low-level contraction of muscle fibers
- C) The maximum contraction of muscle fibers
- D) The lack of muscle activity

Answer: B) The constant, low-level contraction of muscle fibers

115. **Which muscle assists in respiration by elevating the ribs?**

- A) Diaphragm
- B) Pectoralis major
- C) Intercostal muscles
- D) Rectus abdominis

Answer: C) Intercostal muscles

116. **What is the term for a muscle that opposes the action of another muscle?**

- A) Agonist
- B) Antagonist
- C) Synergist
- D) Fixator

Answer: B) Antagonist

117. **What type of muscle contraction is primarily utilized during lifting a weight?**

- A) Isometric contraction
- B) Eccentric contraction
- C) Concentric contraction

- D) Static contraction

Answer: C) Concentric contraction

118. **What is the passive element that provides elasticity to muscle tissue?**

- A) Actin

- B) Myosin

- C) Titin

- D) Tropomyosin

Answer: C) Titin

119. **In which location would you find the iliopsoas muscle?**

- A) In the upper arm

- B) In the thigh

- C) In the abdomen

- D) In the lower leg

Answer: C) In the abdomen

120. **What is the primary function of the gastrocnemius muscle?**

- A) Flex the knee and plantarflex the foot

- B) Extend the knee and dorsiflex the foot

- C) Adduct the thigh

- D) Rotate the hip

Answer: A) Flex the knee and plantarflex the foot

121. **Which muscle is responsible for flexing the forearm at the elbow?**

- A) Biceps brachii

- B) Triceps brachii

- C) Brachialis

- D) Deltoid

Answer: A) Biceps brachii

122. **What type of contraction occurs when the muscle generates force but does not change length?**

- A) Concentric contraction

- B) Eccentric contraction

- C) Isometric contraction

- D) Isotonic contraction

Answer: C) Isometric contraction

123. **Which muscle primarily extends the thigh at the hip joint?**

- A) Iliopsoas

- B) Quadriceps femoris

- C) Gluteus maximus

- D) Hamstrings

Answer: C) Gluteus maximus

124. **What is the role of the diaphragm during inhalation?**

- A) It contracts and moves upward.
- B) It relaxes and allows lung expansion.
- C) It contracts and moves downward, increasing thoracic cavity volume.
- D) It prevents airflow to the lungs.

Answer: C) It contracts and moves downward, increasing thoracic cavity volume.

125. **Which muscle is primarily responsible for the adduction of the arm?**

- A) Latissimus dorsi
- B) Pectoralis major
- C) Deltoid
- D) Rotator cuff muscles

Answer: B) Pectoralis major

126. **What process converts glucose into lactic acid when oxygen levels are low in muscles?**

- A) Aerobic respiration
- B) Anaerobic glycolysis
- C) Citric acid cycle
- D) Gluconeogenesis

Answer: B) Anaerobic glycolysis

127. **Which muscle is known as the "shin muscle" that dorsiflexes the foot?**

- A) Tibialis anterior
- B) Gastrocnemius
- C) Soleus
- D) Peroneus longus

Answer: A) Tibialis anterior

128. **What is the role of creatine phosphate in muscle cells?**

- A) To produce ATP during anaerobic conditions
- B) To provide a quick source of energy for ATP regeneration
- C) To store calcium ions
- D) To transport oxygen

Answer: B) To provide a quick source of energy for ATP regeneration

129. **Which muscle group is involved in the action of sitting up from a lying position?**

- A) Abdominal muscles
- B) Back extensors
- C) Hip flexors
- D) Quadriceps

Answer: A) Abdominal muscles

130. **What is the role of the rotator cuff muscles?**

- A) Stabilize and move the shoulder joint
- B) Extend the elbow
- C) Flex the wrist

- D) Rotate the hip

Answer: A) Stabilize and move the shoulder joint